EXAM 2026

March 2026

1 Introduction

The purpose of this study is to study the performance of two different estimation methods, the Ordinary Least Squares estimator (OLS) and the Feasible Least Squares estimator (FGLS), under the presence of heteroskedasticity in the error term:

$$Var(e_i \mid X_i) = \sigma_i^2$$

2 Data Generating Process

2.1 Generation of regressors

The model is defined as:

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{e}$$

The data in the experiment design have observations each indexed by: $i = 1, \ldots, n$

The DGP contains the following:

- **Regressor** ($\boldsymbol{X}$) To represent the intercept β_0, a column of ones is included in the regressor as: $\boldsymbol{X} = [1,\, X_{1i}]^T$. X_{1i} is defined as a variable drawn from a uniform distribution $X_{1i} \sim U(0, 3)$. This means that any value between 0 and 3 can occur, allowing for a wide range of intensity for the heteroskedasticity in the experiment.
- **Parameters** ($\boldsymbol{\beta}$) The true values of $\boldsymbol{\beta}$ are set to $\boldsymbol{\beta} = [\beta_0,\, \beta_1]^T = [3, 0.9]^T$. These values are chosen as the true values of $\boldsymbol{\beta}$ to compare how "correct" the estimators are relative to the true value. Furthermore, to be able to evaluate the estimators, the true value has to be known.
- **Error term** ($\boldsymbol{e}$) The error terms are drawn from the following distribution:

  $$e_i \sim N(0, e^{x_{1i}\lambda})$$

  The error terms are independent, drawn from a normal distribution with mean 0. However, the error terms are heteroskedastic in an exponential form. The lambda parameter is defined as $\lambda = 2.5$

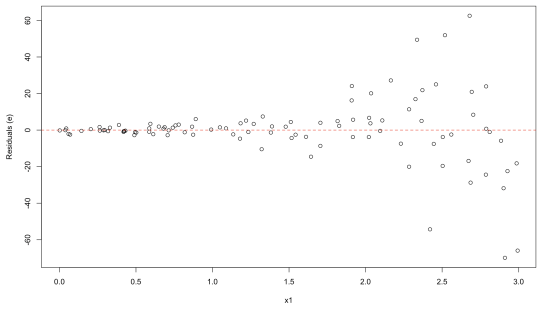


Figure 1: Residuals (e_i) against X_{1i} illustrating exponential heteroskedasticity.

The structure of the heteroskedasticity can be seen clearly in Figure 1, when plotting X_{1i} against e_i, illustrating the exponential form.

The data-generating process relies on assumptions necessary for linear regression. Firstly, it relies on assumption 4.1, that the random variables $((Y_1, X_1), ...(Y_i.X_i), ...(Y_n, X_n))$ are independent and identically distributed, which is fulfilled through our DGP (Hansen 2022, p. 98).

Then it relies on assumption 4.2, which includes several assumptions (Hansen 2022, page 100):

1. Firstly, the variables (Y, X) satisfy the linear regression equation:

$$Y = X\beta + e$$

In our simulation design, the linear regression equation is satisfied through a linear model in our data-generating process for the experiment.

2. The following assumption is also needed:

$$E[e \mid X] = 0$$

Meaning that the conditional mean of the errors is 0. This implies exogeneity, meaning X gives us no information about the error; they are not correlated at all. This will also lead to the estimator being unbiased:

$$E[\hat{\beta}] = \beta$$

Since the errors are drawn from a normal distribution with mean 0, this assumption will also be fulfilled.

3. Then, the assumption of finite second moments is needed:

$$E[Y^2] < \infty$$
$$E \mid\mid X \mid\mid^2 < \infty$$

The variables have finite second moments since X_{1i} is drawn from a bounded uniform distribution, and errors are generated from a normal distribution with mean 0 and variance $e^{x_{1i}\lambda}$. Since X and the errors are finite, Y will be guaranteed to be finite too, therefore fulfilling the assumptions.

4. An invertible design matrix is required:

$$\boldsymbol{Q}_{XX} = E[XX^T] > 0$$

For X to be invertible, it is required to have full column rank. Since X_{1i} is generated from the uniform distribution with the interval (0, 3), it is guaranteed to have $Var(X_{1i}) > 0$, therefore not being a linear combination of the intercept, therefore having full rank, and leading to the design matrix being an invertible matrix.

5. Finally, homoskedasticity is needed, meaning:

$$E[e^2 \mid X] = \sigma^2(X) = \sigma^2$$

Thus, the only assumption not being fulfilled is the assumption of homoskedasticity, meaning our experiment design allows us to evaluate the effect of the estimators under heteroskedasticity, without any other assumption being violated.

The sample size for the single DGP was chosen to be $n = 100$. The sample size was chosen to serve as a good comparison between the estimators, not too large where the difference between the estimators would not be seen, but not too small where the estimators would be too unstable.

3 Ordinary least squares estimator

3.1 Calculating the least squares estimator and variance

Now, we implement the ordinary least squares estimator (OLS) using the formula (Hansen 2022, p. 101):

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

where:

- $\hat{\boldsymbol{\beta}}$ is a (2 x 1) vector with estimated coefficients
- $\boldsymbol{X}$ is a (100 x 2) matrix. It is the design matrix with the explanatory variables and a column of ones for the intercept
- $\boldsymbol{Y}$ is a (100 x 1) column vector containing the dependent variable

Since heteroskedasticity is present in the data, the classic covariance matrix estimator tends to have a high amount of bias, which will be explained upon further later in the report. Therefore, we introduce a heteroskedasticity-consistent covariance matrix estimator. According to Hansen, one often recommended approach under heteroskedasticity is to use the HC1 estimator (Hansen 2022, p. 114):

$$\hat{\boldsymbol{V}_{\boldsymbol{\beta}}}^{HC1} = \frac{n}{n-k}(\boldsymbol{X^T X})^{-1}(\sum_{i=1}^{n} X_i X_i^T \hat{e}_i^2)(\boldsymbol{X^T X})^{-1}$$

To obtain the residuals $\hat{e}_i$ the model is estimated using the OLS regression:

$$\hat{e}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$$

Note: In the implementation in the code, this is performed in matrix form $\hat{\boldsymbol{e}} = \boldsymbol{Y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}$ to ensure efficiency in the calculations

The middle part $X_i X_i^T \hat{e}_i^2$, often called "the meat" of the sandwich estimator, is calculated by weighing each observation by its squared residuals and then summing them:

$$\text{Meat} = \sum_{i=1}^{n} X_i X_i^T \hat{e}_i^2$$

The "bread" part on both sides $(\boldsymbol{X^T X})^{-1}$ is then multiplied, and then finally everything was multiplied by the adjustment term $\frac{n}{n-k}$, thus giving us the heteroskedasticity-consistent covariance matrix estimator.

3.2 The problem with the usual homoskedastic variance formula

The usual homoskedastic variance formula is defined to be:

$$\boldsymbol{V}_{\hat{\beta}} = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

This formula relies on several assumptions. Firstly, it relies on the previous assumption 4.1 and on assumptions 4.2. Homoskedasticity is also needed. The assumption that is violated here is the assumption of homoskedasticity (Hansen 2022, p. 103). Under homoskedasticity, we have the following case (Hansen 2022, p 103):

$$\boldsymbol{D} = \boldsymbol{I}_n\sigma^2 = \begin{pmatrix} \sigma^2 & 0 & ... & 0 \\ 0 & \sigma^2 & ... & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & ... & \sigma^2 \end{pmatrix}$$

This allows us to do the following simplifications for the variance formula (Hansen 2022, p.103):

$$\boldsymbol{X}^T\boldsymbol{D}\boldsymbol{X} = \boldsymbol{X}^T\boldsymbol{X}\sigma^2$$

$$\boldsymbol{V}_{\hat{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}(\boldsymbol{X}^T\boldsymbol{D}\boldsymbol{X})(\boldsymbol{X}^T\boldsymbol{X})^{-1} = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

Thus giving us the usual homoskedastic formula. From this, the classic method to estimate σ^2 is through the unbiased estimator (Hansen 2022, p. 110):

$$s^2 = \frac{1}{n-k}\sum_{i=1}^{n}\hat{e_i}^2$$

However, in our situation under heteroskedasticity, the covariance matrix will look like the following instead:

$$\boldsymbol{D} = diag(\sigma_1^2, ..., \sigma_n^2) = \begin{pmatrix} \sigma_1^2 & 0 & ... & 0 \\ 0 & \sigma_2^2 & ... & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & ... & \sigma_n^2 \end{pmatrix}$$

The homoskedasticity assumption is therefore violated. Every observation will not have equal variance since it is now observation-specific. The simplification $\boldsymbol{D} = \boldsymbol{I}_n\sigma^2$ will not be correct and accurate anymore, making the usual homoskedastic formula for the variance be inaccurate. Every observation will be treated as having the same error variance σ^2 even when this is no longer the case under heteroskedasticity; some observations will have a much greater variance than others. This will lead to biased standard errors, which is the reason we instead prefer the HC1 estimator.

By using the HC1 estimator, instead of assuming equal σ^2 the HC1 estimator uses each observation's unique squared error term estimate $\hat{e_i}^2$, acknowledging that observations will have different amounts of variance. Since $\hat{e}_i$ tends to be biased towards zero, it is also scaled by $\frac{n}{n-k}$, providing more accurate standard errors, derived from the more correct variance estimator (Hansen 2022, p. 114).

3.3 Results

Table 1: Results from OLS regression

	True Beta	Estimate	Std_Error
β_0	3	1.409	2.45
β_1	0.9	2.456	2.75

Looking at Table 1, we notice that our estimates are a bit off, but not entirely wrong. Even if OLS remains unbiased, it is no longer efficient (Hansen 2022, p. 108). The smaller sample size of n = 100 and the heavy heteroskedasticity cause the OLS estimate to be imprecise, caused by the high sampling variance, leading to the difference between the true values and the estimates. The standard errors that were calculated using HC1 (robust standard errors) are very high, for β_0, around 2.45, while β_1 has a standard error around 2.75, showing substantial uncertainty.

Recall the formula for the least squares estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Note that the least squares formula treats every observation equally, causing the variance to be high and estimates to be off due to the sampling variance. No weight matrix is used in the estimate. Since all observations will have equal weights, even those with much higher variance, this will cause the estimates to be imprecise and the standard errors to be high.

4 Feasible GLS Estimation

Now we turn to GLS estimation. When we have heteroskedasticity, OLS will no longer provide the lowest variance (Hansen 2022, p. 108). According to Theorem 4.5, the lower bound for the variance is given by the following statement:

If $\tilde{\beta}$ is an unbiased estimator of β then:

$$var[\tilde{\beta} \mid \boldsymbol{X}] \geq \sigma^2(\boldsymbol{X}^T\Sigma^{-1}\boldsymbol{X})^{-1}$$

GLS will reach this lower bound where $\Sigma = \boldsymbol{D} = diag(\sigma_1^2, ...\sigma_n^2)$ (where we have heteroskedasticity) (Hansen, 2022 p. 108). GLS is then given by:

$$\boldsymbol{\beta_{GLS}} = (\boldsymbol{X}^T\boldsymbol{D}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{D}^{-1}\boldsymbol{Y}$$

Since in this exercise, we do not know $\boldsymbol{\Sigma}$, we use an estimate; thus, it becomes feasible GLS (Hansen 2022, p. 108).

4.1 Estimation

First, we have to assume a structure to the variance. Here, we assume an exponential form of the variance in the following regression form:

$$\sigma_i^2 = e^{x_{1i}\lambda}$$

λ represents the unknown scaling factor that tells how much the error is expected to fluctuate.

We then need an estimate for σ_i^2. To get this, we run the original OLS regression on the original model and obtain the residuals $\hat{e_i}$. We then use the squared residuals ($\hat{e_i}^2$) as the estimator for σ_i^2, representing our best guess of our estimate. Since OLS can only be done for linear relationships, we convert the exponential relationship to a linear:

$$ln(\hat{e}_i^2) = X_{1i}\lambda + v$$

We then use the squared residuals to try to estimate the unknown scaling parameter λ using OLS.

$$\hat{\lambda} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\ln(\hat{\boldsymbol{e}}^2)$$

We then calculate the variance estimate $\hat{\sigma}_i^2$, based on our estimates:

$$\hat{\sigma}_i^2 = \hat{e_i}^2 = e^{x_{1i}\hat{\lambda}}$$

We can now calculate the weight matrix, which is the inverse of the variance-covariance matrix.

$$\boldsymbol{D}^{-1} = diag(\frac{1}{\hat{\sigma}_1^2}, \frac{1}{\hat{\sigma}_2^2}..., \frac{1}{\hat{\sigma}_n^2})$$

Finally, we run the regression with the estimated $\boldsymbol{D}^{-1}$ to get our estimates, giving us the feasible generalized least squares regression (FGLS):

$$\hat{\beta}_{FGLS} = (\boldsymbol{X}^T\boldsymbol{D}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{D}^{-1}\boldsymbol{Y}$$

4.2 Results

Table 2: Results from FGLS estimation

	True Beta	**Estimate**	**Std_Error**
β_0	3	2.752	0.337
β_1	0.9	1.046	0.586

Analyzing the results from the feasible GLS estimation in Table 2 shows a substantial improvement. Just like the OLS estimator, the GLS estimator is also unbiased. However, it can be seen how the FGLS estimator is much more efficient, and the estimates are closer to the true values. The standard errors have also diminished dramatically, being much smaller than the errors for the OLS estimate.

This illustrates how by adding weights through $\frac{1}{\hat{\sigma}_i^2}$ improves our estimation drastically. The reason FWLS achieves a lower variance than the OLS estimation is because of the heavy heteroskedastic structure in the error terms. The variance in our data grows exponentially through $e^{x_{1i}\times 2.5}$. The OLS formula lacks any form of weight matrix and treats every observation equally, assigning a weight of 1 to each point regardless of the reliability in the form of its variance, which will result in the estimator being less precise and more unreliable. This is why feasible generalized least squares (FGLS) is better. Instead of treating each observation equally, observations are given weights through the assumed error structure, providing a lower standard error and a more precise estimate than the regular OLS.

5 Monte Carlo Comparison

5.1 Measurements

Monte Carlo estimation allows us to evaluate the performance of a statistical procedure in a more realistic environment (Hansen 2022, p. 242). This is done by performing a large number of replications; in each replication, we draw the chosen sample size. By doing this, it allows us to further analyze the OLS estimation in comparison to the FGLS estimation. The sample size was set to 100, like the previous estimations, but the experiment is now repeated 1000 times for both the OLS and FGLS estimations. To compare the estimators, the empirical bias was first calculated, using the following formula (Hansen 2022, p.242):

$$\text{Emphirical Bias} = \frac{1}{M}\sum_{m=1}^{M}(\widehat{\beta}_m - \beta)$$

where

- M is the total number of replications
- $\widehat{\beta}_m$ is the β estimate for m:th replication
- β is the true value

Then, the empirical variance was calculated using the following formula (Hansen 2022, p.242):

$$Var(\widehat{\beta}) = \frac{1}{M-1}\sum_{m=1}^{M}(\widehat{\beta_m} - \widehat{\beta})^2$$

where

- M is the total number of Monte Carlo replications
- $\widehat{\beta}_m$ is the estimate for the m:th replication
- $\widehat{\beta}$ is the mean of the estimates

Finally, the mean squared error (MSE) was calculated using the following formula (Hansen 2022, p.242):

$$MSE(\hat{\beta}) = (\widehat{\text{bias}[\hat{\beta}]})^2 + Var(\hat{\beta})$$

5.2 Results

Performing the Monte Carlo simulation with sample size n = 100 and 1000 replications gave the following results:

Table 3: Monte Carlo simulation n = 100

	Bias	**Variance**	**MSE**
OLS_{β_0}	-0.024	4.8	4.8
OLS_{β_1}	0.05	5.438	5.438
$FGLS_{\beta_0}$	0.003	0.194	0.194
$FGLS_{\beta_1}$	0.0127	0.625	0.625

Looking at the table, it can be seen how the FGLS now manages to perform substantially better compared to OLS. Both estimators have a bias close to 0, illustrating that they are unbiased. However, it can now be seen how FGLS manages to achieve a drastically lower variance and MSE for both β values compared to the OLS estimator.

The MSE will be very close to the variance estimates since it is calculated with the bias, which is very close to 0 here. Just like the results for the single FGLS estimation, this illustrates that FGLS will be able to perform more efficiently and achieve lower variance by weighting the observations instead of giving equal weights. OLS illustrates higher variance and MSE, caused by treating every observation equally, even though this will not be the case under the exponential heteroskedastic structure, with observations with a large x_{1i}value having a much higher variance, causing noisier observations and a higher sampling variance than FGLS, which accounts for this.

By the Gauss-Markov theorem, under the assumption of the linear regression model, $E[e \mid X] = 0$ and the homoskedasticity assumption, OLS is the **best linear unbiased estimator (BLUE)**. This means that if $\tilde{\beta}$ is an unbiased estimator of β, the following holds (Hansen 2022, p.105):

$$var[\tilde{\beta} \mid \boldsymbol{X}] \geq \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

The results from the FGLS estimation clearly illustrate how, when the homoskedasticity assumption is violated, this is no longer the case, showing that the FGLS estimator now provides lower variance and MSE, and lower standard deviation, which could be seen from the single-run FGLS.

Table 4: Standard Deviation comparison

Estimator	**Single Run Standard Deviation**	**MC Standard Error**
OLS (β_0)	2,45	2,19
FGLS (β_0)	0,337	0,44
Estimator	**Single Run Standard Deviation**	**MC Standard Error**
OLS (β_1)	2,75	2,33
FGLS (β_1)	0,586	0,79

From the Monte Carlo estimation, it can also be seen how the standard deviations align closely to the single run standard deviations for all estimates for both methods, suggesting that our single run estimation in the previous results is representative and a realistic measure of how the estimator performs.

6 Comparison of sample sizes

Finally, we compare the Monte Carlo estimation results using a bigger sample size, n = 1000, with 1000 replications.

Table 5: Monte Carlo simulation n = 1000

	Estimator	**Bias**	**Variance**	**MSE**
β_0	**OLS**	-0.002	0.5	0.5
β_0	**FGLS**	0.006	0.0147	0.0147
β_1	**OLS**	0.0011	0.592	0.592
β_1	**FGLS**	-0.007	0.048	0.048

Analyzing the bias of β_0 and β_1 the estimators can be seen to still, like in the small sample, have a very small bias, illustrating that they are both unbiased estimators.

FGLS still beats OLS in lower variance and MSE, showing that the assumption of the heteroskedastic structure and assigning weights still results in more accurate and stable estimates for both sample sizes, as opposed to OLS, which treats every observation equally and thus remains inefficient.

However, when increasing the sample size to 1000, both variance and MSE decrease drastically for both estimators, making both estimators have very small variances and MSE, confirming that when we have more data, the variance will be lower. This illustrates that the asymptotic properties of both estimators, OLS and the FGLS estimator, have consistency, with a variance converging towards 0 when the sample size grows. While the FGLS consistently performs better, it can still be seen that OLS also serves as a consistent estimator in large sample sizes when the variance decreases substantially.

References

Hansen, B. E. (2023) *Econometrics*. Princeton: Princeton University Press.